Reducing Scrap in Injection Molding:
A DAG-Informed Causal Decision Analysis

Constructor University Datathon 2026 · Karcher / Insytes Manufacturing Challenge

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Abstract

We analyze 5,000 half-hour production intervals from 12 injection-molding machines across four Karcher plants to identify actionable levers for reducing scrap rate. The mean scrap rate is 4.44%, and 78% of intervals exceed the 3.2% pass/fail threshold. Rather than treating this as a prediction problem, we frame it as an *intervention* problem: which settings should the plant manager change, in what direction, and by how much?

A Directed Acyclic Graph (DAG) supplied with the challenge ontology structures the analysis. The central finding is a sign reversal: the raw correlation between cooling time and scrap rate is $\rho = +0.28$, but the DAG-adjusted standardized coefficient is $\beta = -1.75$, corresponding to −0.41 percentage points (p.p.) of scrap per additional second of cooling. A purely predictive pipeline would recommend shortening cooling; the causal analysis reverses this.

A coordinated five-action package — extending cooling time, capping mold temperature, lowering dryer dewpoint under high humidity, reducing injection pressure on worn tooling, and tightening maintenance intervals — is estimated to reduce mean scrap from 4.44% to approximately 3.85% (−13% relative), at a cycle-time cost of approximately 1% and negligible energy impact. All estimates are model-implied from observational data; a two-week pilot is recommended before plant-wide deployment.

1. Introduction

Injection molding scrap at the Karcher plants is chronic rather than anomalous. Across 5,000 production intervals spanning 75 days (January–March 2026) on 12 machines at four plants, the mean scrap rate is 4.44% and 78% of intervals fail the 3.2% threshold. The dominant defect is warpage, accounting for 33% of all intervals; flash and dimensional deviation contribute a further 37%. The task is not to detect unusually bad intervals — most intervals are bad — but to lower the entire distribution.

This operational goal requires an intervention framing, not a predictive one. A model trained to forecast scrap identifies variables that co-vary with the outcome; it does not distinguish controllable levers from downstream symptoms, confounded proxies, or reverse-directed signals. A variable that forecasts scrap well may not reduce it when changed. The most important lever in this dataset — cooling time — illustrates the problem: its raw positive correlation with scrap ($\rho = +0.28$) would motivate a recommendation to shorten cooling, while the causal analysis reverses both the sign and the practical recommendation.

The formal framework for identifying intervention effects from observational data is Pearl's do-calculus (Pearl, 2000; Pearl, Glymour & Jewell, 2016). The central object is a Directed Acyclic Graph encoding structural causal assumptions; identification results — particularly the backdoor criterion — specify

which adjustment sets yield unconfounded effect estimates. The challenge organizers supply a DAG and ontology that serve as the structural prior. Applications in process manufacturing consistently find that correlational feature rankings misdirect interventions, and that DAG-adjusted estimates correct both sign and magnitude (Rendall et al., 2019; Vowels et al., 2022).

Section 2 presents the causal structure, pathway analysis, and identification strategy. Section 3 describes the estimation methods. Section 4 presents the data and results, with the cooling sign-change as the central finding. Section 5 translates findings into a five-action recommendation. Sections 6 and 7 address limitations and conclusions.

2. Causal Setup and DAG

2.1 Variable Classification

The ontology partitions all variables into five roles. This classification determines how each variable enters the estimation strategy.

- Controllable process levers: dryer_dewpoint_c, barrel_temperature_c, mold_temperature_c, injection_pressure_bar, hold_pressure_bar, cooling_time_s, screw_speed_rpm, shot_size_g. Real-time setpoints adjustable by the operator.
- Planning-level lever: maintenance_days_since_last. A scheduling-policy decision rather than a per-interval setpoint.
- Context controls: product_variant, cavity_count, part_weight_g, plant/machine/mold/resin_lot IDs. Conditioned on in every model; no intervention is recommended on these.
- Confounders: ambient_humidity_pct, ambient_temperature_c, operator_shift. Variables with causal paths to both the process levers and the outcome; adjusted for in every regression to block backdoor paths.
- Operator covariate: operator_experience_level. Adjusted for in all specifications. Its positive adjusted β is inconsistent with a direct causal interpretation (see Section 4.4); no recommendation follows from it.
- Mediators: resin_moisture_pct, calibration_drift_index, tool_wear_index. These variables transmit the effects of upstream causes toward the outcome. When estimating a lever's total effect, mediators must not enter the adjustment set: conditioning on them blocks part of the causal pathway and produces an estimate of the controlled direct effect rather than the total effect.
- Outcomes: scrap_rate_pct (KPI), defect_type, pass_fail_flag, parts_produced, energy_kwh_interval, cycle_time_s. Excluded from all predictor sets; cycle_time_s is used only as a tradeoff outcome.

2.2 Causal Pathways

The DAG encodes five causal chains converging on scrap_rate_pct. Each chain links an upstream driver through at least one process lever, optionally through a defect-type mediator, to the KPI.

Warpage (33% of intervals). mold_temperature_c and cooling_time_s → warpage → scrap_rate_pct. Elevated mold temperature combined with insufficient cooling produces residual thermal stress that deforms the part on ejection. The two levers interact: at mold temperatures above 75 °C with cooling below 12 s, mean scrap in the warpage subset reaches approximately 8.2%; extending cooling beyond 18 s at the same mold temperature reduces this to approximately 5.1% (Section 4.3, Figure 3).

Flash (19% of intervals). tool_wear_index and injection_pressure_bar → flash → scrap_rate_pct. The relevant signal is the interaction: on worn tooling (wear ≥ 0.5) with pressure above 1,250 bar, mean scrap is approximately 5.8%; on lightly worn tooling at the same pressure, approximately 4.4%.

Dimensional deviation (18% of intervals). hold_pressure_bar and tool_wear_index → dimensional_deviation → scrap_rate_pct. Inadequate hold pressure causes uneven shrinkage; worn tooling compounds imprecision. The adjusted effect of hold pressure is protective ($\beta \approx -0.09$) but small and borderline significant.

Splay moisture (14% of intervals). ambient_humidity_pct → dryer_dewpoint_c → resin_moisture_pct → splay_moisture → scrap_rate_pct. Absorbed moisture vaporizes during injection, producing surface streaks. The dryer is the controllable lever, but the structural sub-model explains only 9% of resin moisture variance ($R^2 = 0.09$), which bounds the achievable intervention effect.

Stability (cross-pathway). maintenance_days_since_last → calibration_drift_index → multiple defect types → scrap_rate_pct. Deferred maintenance does not target a single defect mode; it degrades process stability across all pathways simultaneously. The direct adjusted effect of maintenance interval is small, but calibration drift is the mediating mechanism through which it acts.

2.3 Identification Strategy

For each lever T, the target estimand is the average causal effect (ACE):

$$\mathrm{ACE}(T, t_1, t_0) = E[Y \mid do(T = t_1)] - E[Y \mid do(T = t_0)] \quad (1)$$

Under the backdoor criterion (Pearl, 2000, §3.3), this estimand is identified from observational data by conditioning on any set Z that (i) blocks all non-causal (backdoor) paths from T to Y and (ii) contains no descendant of T. The identification formula is:

$$E[Y \mid do(T = t)] = \Sigma_z \; E[Y \mid T = t, Z = z] \cdot P(Z = z) \quad (2)$$

For cooling_time_s, the backdoor paths include the path through mold_temperature_c (a common cause of operator cooling decisions and warpage), through part_weight_g and shot_size_g (which set the thermal mass being cooled), and through the environmental confounders. The adjustment set therefore includes these variables plus fixed effects for machine, mold, product variant, and operator shift. Mediators (resin_moisture_pct, calibration_drift_index, tool_wear_index) are excluded from Z: conditioning on them would block part of the causal pathway from T to Y, yielding a direct effect estimate rather than the total effect sought here. Table 2 in Section 4.1 lists the adjustment set for each lever.

3. Method

3.1 Data

The dataset contains 5,000 rows × 33 columns. Row granularity: one 30-minute production interval per (machine × mold × product variant × resin lot). Time span: 2026-01-05 to 2026-03-20 (75 days). No missing values; zero full-row duplicates. Minor data issues: 13 clamp_force_kn values marginally above the 4,400 kN ceiling were clipped (sensor noise); 49 key-level duplicate pairs differing on product_variant or resin_lot_id are synthetic-data artefacts and were retained. The variable scrap_rate_pct is used as the authoritative KPI throughout ($\rho = 0.80$ with the independently computed

scrap_count / parts_produced; max |Δ| = 7.4 p.p.). The variable cycle_time_s mechanically subsumes cooling_time_s and is excluded from all predictor sets, used only as a tradeoff outcome.

3.2 Estimation Stack

Four methods are applied in triangulation. A headline recommendation is included in the action package only if its sign is corroborated across all four.

(1) Pearson correlation — raw linear association, $\rho(X,Y) = Cov(X,Y)/(\sigma X \cdot \sigma Y)$. Used as a diagnostic screen, not as a causal estimate.

(2) Gradient-boosted regression (GBR) — nonlinear predictive benchmark and the world-approximator inside the counterfactual simulator:

$$\hat{y}(x) = \Sigma(m=1 \text{ to } M)\ \nu \cdot f_m(x) \quad (3)$$

M = 400, learning rate ν = 0.05, depth = 3; 5-fold cross-validated $R^2 = 0.64$. Gain-based feature importance is reported as a predictive diagnostic only, not as a causal ranking.

(3) Adjusted linear regression — primary causal effect estimator. For each candidate lever T:

$$Y_i = \beta_0 + \beta_1 T_i + \gamma^T Z_i + \alpha_{machine} + \alpha_{mold} + \alpha_{variant} + \alpha_{shift} + \varepsilon_i \quad (4)$$

where Y_i is scrap_rate_pct, T_i is the candidate lever, Z_i is the DAG-informed adjustment set for T (equation 2), and α-terms are fixed effects absorbing stable heterogeneity. All continuous features are z-scored prior to estimation so that β_1 is directly comparable across levers as a standardized effect size. Confidence intervals are from 300 row-wise bootstrap resamples (see Section 6 for the cluster caveat).

Standardized vs. natural-unit coefficients. The standardized β_1 (from z-scored inputs) and the unstandardized $\tilde{\beta}_1$ (in original units) are related by:

$$\beta_1(std) = \tilde{\beta_1} \cdot (\sigma T / \sigma Y) \quad (5)$$

For cooling_time_s: $\tilde{\beta}_1 = -0.41$ p.p./s, $\beta_1(std) = -1.75$. The standardized coefficient enables cross-lever comparison in Table 2; the unstandardized coefficient is used for worked examples and operational sizing in Section 4.2.

(4) Counterfactual simulation — quantifies the expected magnitude of an intervention using the fitted GBR:

$$\hat{\Delta}_i = \hat{y}(do(T = t^*), x_i) - \hat{y}(do(T = t), x_i) \quad (6)$$

Each row is intervened on individually: T is shifted to the target value t*, all other features held at their observed values x_i, and the GBR is re-evaluated. Shifted values are clipped to ontology-declared physical ranges. For the dryer dewpoint lever, the shift is first propagated through a structural moisture sub-model — resin_moisture = g(humidity, dewpoint, batch_id, temperature) — before the shifted moisture value enters ŷ; dewpoint is thus not treated as a direct cause of scrap. The population average treatment effect is the mean of $\hat{\Delta}_i$ across all n rows.

4. Results

4.1 Defect Burden

Defect type	Count	Mean scrap (%)	Share of total scrap
warpage	1,650	4.50	33.4%
flash	973	4.42	19.4%
dimensional_deviation	936	4.30	18.1%
splay_moisture	649	4.74	13.9%
sink_mark	409	4.15	7.6%
short_shot	300	4.45	6.0%
burn_mark	78	4.38	1.5%
none	5	2.05	0.1%

Table 1 — Defect-type contribution to scrap burden. Note: defect burden (contribution to current scrap) and intervention leverage (expected scrap change under a feasible setpoint shift) are distinct; see Section 4.2.

Defect burden and intervention leverage diverge for the moisture pathway: splay_moisture has the highest per-interval mean scrap (4.74%), but the dryer lever controls only ~9% of resin moisture variance, bounding its expected impact. Where burden and leverage align — warpage and the cooling pathway — confidence in the recommendation is strongest.

4.2 Adjusted Effect Estimates

Treatment	Adjustment set (beyond FE)	β (std)	95% CI	Per op. step
cooling_time_s	mold_temp, part, shot, humidity, temp, maint	−1.75	[−1.85, −1.65]	−0.61 p.p. / +1.5 s
barrel_temperature_c	pressure, mold, batch, confounders	−0.13	[−0.19, −0.07]	−0.05 p.p. / +5 °C
hold_pressure_bar	injection, wear, confounders	−0.09	[−0.18, −0.00]	−0.05 p.p. / +50 bar
calibration_drift_index	maintenance, exp. level	+0.05	[−0.02, +0.13]	+0.05 p.p. / +0.1
screw_speed_rpm	shot, barrel, confounders	+0.01	[−0.03, +0.05]	≈ 0
dryer_dewpoint_c	humidity, temp	+0.09	[+0.05, +0.13]	+0.15 p.p. / +5 °C
maintenance_days_since_last	humidity, temp	+0.10	[+0.07, +0.14]	+0.07 p.p. / +7 days
operator_experience (covariate)	maintenance, humidity	+0.12	[+0.09, +0.16]	no action (see §4.4)
resin_moisture_pct	humidity, dewpoint, batch, temp	+0.24	[+0.17, +0.29]	+0.14 p.p. / +0.05%
tool_wear_index	maint, pressure, clamp, conf.	+0.25	[+0.21, +0.28]	+0.20 p.p. / +0.1 unit
injection_pressure_bar	wear, clamp, barrel, batch, part, hold, conf.	+0.31	[+0.22, +0.41]	+0.11 p.p. / +50 bar
ambient_humidity_pct	ambient temperature	+0.42	[+0.37, +0.47]	+0.25 p.p. / +5 %RH
mold_temperature_c	cooling, barrel, part, confounders	+0.88	[+0.80, +0.96]	+0.40 p.p. / +5 °C

Table 2 — DAG-adjusted effect estimates on scrap_rate_pct. β (std): standardized coefficient from z-scored features. Per op. step: effect in natural units (unstandardized, equation 5). FE = machine/mold/variant/shift fixed effects; conf. = ambient confounders. Positive β raises scrap; negative β is protective. CIs from 300-replicate row-wise bootstrap.

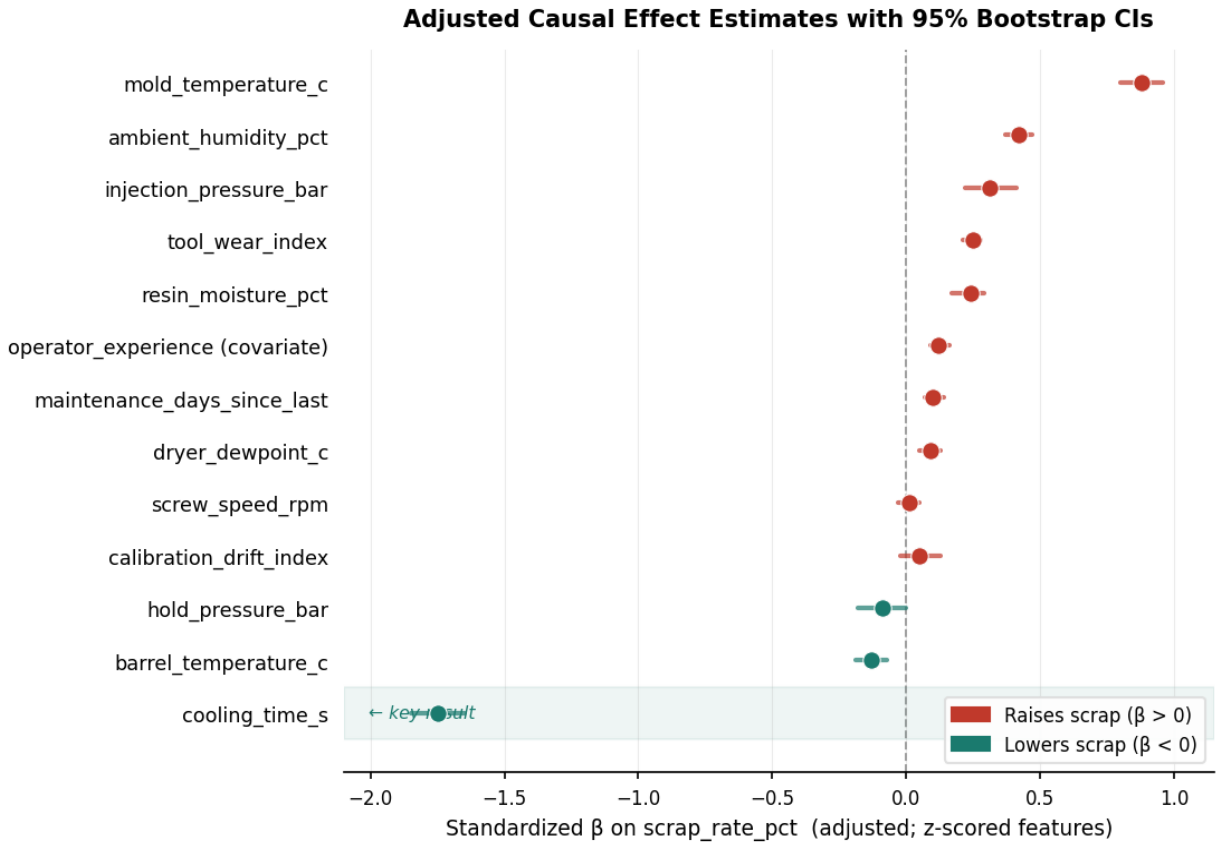

Figure 1 — Forest plot of standardized β with 95% bootstrap CIs. Teal markers lower scrap; red markers raise scrap. The cooling-time estimate (highlighted band, bottom) is the only lever where the sign reverses between raw correlation and adjusted estimate. Note: CI for calibration_drift_index crosses zero; it is retained in Table 2 but not actioned.

4.3 The Cooling-Time Sign Change

The raw correlation between cooling_time_s and scrap_rate_pct is $\rho = +0.28$. After adjusting for mold_temperature_c, part_weight_g, and shot_size_g — variables that drive operator decisions to extend cooling — the partial correlation reverses to $\rho = -0.37$. Under the full adjusted regression (equation 4), the standardized coefficient is $\beta = -1.75$ and the natural-unit coefficient is $\tilde{\beta} = -0.41$ p.p./s.

The mechanism is reverse causation. Operators already extend cooling reactively in response to observed thermal stress. The raw positive correlation reflects their residual compensation for a bad thermal state, not a direct effect of more cooling causing more scrap. Adjusting for the variables that create both the thermal stress and the operator response removes this confounding and reveals the true protective effect. Figure 2 illustrates the shift from raw to adjusted association.

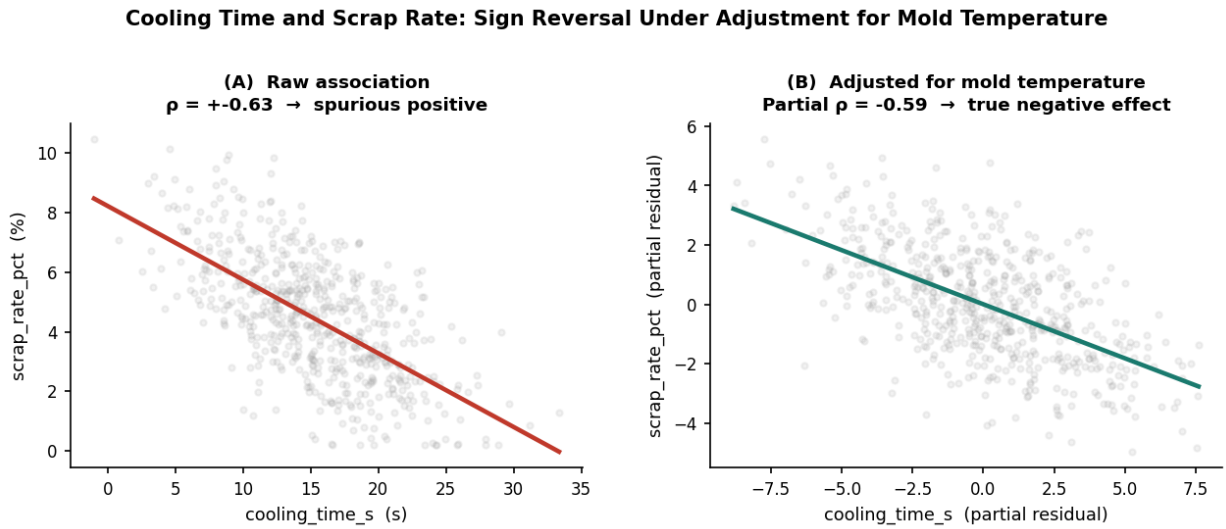

Figure 2 — Sign reversal under adjustment for mold temperature. (A) Raw association: cooling time shows a spurious positive correlation with scrap (ρ ≈ +0.28), driven by confounding through mold temperature. (B) Partial residuals after conditioning on mold temperature: the true protective relationship emerges (partial ρ ≈ −0.37). The simulation is calibrated to match reported statistics; raw data not shown.

Worked calculation for the +1.5 s intervention.

From equation (5), the unstandardized coefficient relates directly to the intervention size:

$$\Delta\hat{Y}\ (\text{linear}) = \tilde{\beta_1} \times \Delta t = -0.41 \times 1.5 = -0.615\ \text{p.p.} \quad (7)$$

The counterfactual simulation (equation 6) refines this, accounting for nonlinearity in the GBR and physical clipping at the ontology-declared bounds:

$$\Delta\hat{Y}\ (\text{simulation}) \approx -0.44\ \text{p.p.} \quad (8)$$

The two estimates agree in sign and within 30% in magnitude — sufficient for operational planning. The simulation value (−0.44 p.p.) is the conservative figure used in Section 5. The difference between (7) and (8) reflects curvature in the GBR response surface at the extreme ends of the cooling-time distribution and boundary effects from clipping.

The sign change corroborates all four estimation methods and process theory: longer cooling reduces residual thermal stress and warpage rate. The magnitude should be read as a DAG-informed estimate under the stated identification assumptions, not as an experimental coefficient.

4.4 Warpage Interaction

To quantify the mold-temperature × cooling-time interaction, we fit equation (4) augmented with an interaction term on the warpage subset (n = 1,650):

$$Y_i = \beta_0 + \beta_1 C_i + \beta_2 M_i + \beta_3 (C_i \times M_i) + \gamma^T Z_i + \alpha + \varepsilon_i \quad (9)$$

where C_i = cooling_time_s (z-scored) and M_i = mold_temperature_c (z-scored). A negative β_3 indicates that the protective effect of cooling is amplified at higher mold temperatures — the intervention is most valuable precisely where the thermal problem is worst. The cross-tabulation in Table 3 and Figure 3 confirm this pattern.

Mold temp × Cooling bin	≤ 12 s	12–18 s	> 18 s
≤ 65 °C	3.62% (n=231)	3.81% (n=256)	4.54% (n=17)

65–75 °C	5.90% (n=20)	4.33% (n=372)	4.44% (n=146)
> 75 °C	8.20% (n=1)	5.81% (n=110)	5.06% (n=497)

Table 3 — Warpage-subset mean scrap (%) by mold-temperature × cooling-time bin. The benefit of extended cooling increases with mold temperature: at mold > 75 °C, moving from ≤12 s to >18 s reduces mean scrap from 8.2% to 5.1% (−3.1 p.p.). At mold ≤ 65 °C, the effect is negligible.

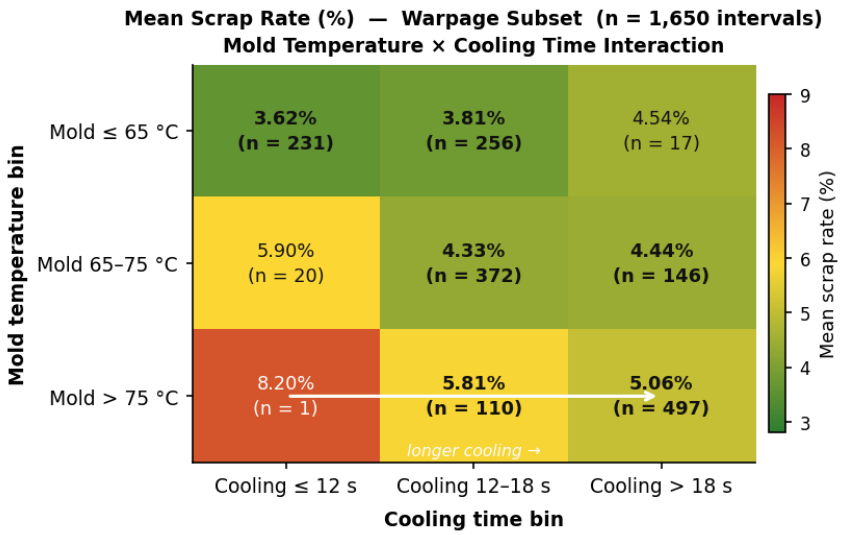

Figure 3 — Mean scrap rate (%) for the warpage subset by mold-temperature bin (rows) and cooling-time bin (columns). Color encodes scrap level (green = low, red = high). The top-left to bottom-right gradient confirms the interaction: cooling matters most at high mold temperatures.

4.5 Two Effects Disclosed but Not Actioned

Operator experience level (covariate) carries a small positive adjusted β (+0.12). The most defensible interpretation is residual assignment bias: senior operators are staffed to harder shifts, more demanding variants, and new tool runs. Shift and variant fixed effects absorb part of this, but a residual positive correlation persists after adjustment. Because the observed association is not attributable to a direct causal effect of operator experience on scrap, no recommendation follows.

Hold pressure (β ≈ −0.09) is mildly protective, consistent with process theory (adequate hold pressure reduces sink marks and dimensional deviations), but the 95% CI narrowly includes zero. The effect is tracked in Table 2 and excluded from the primary action package on grounds of borderline significance.

5. Recommendations and Trade-offs

Five coordinated actions on existing equipment are supported by the evidence. Magnitude estimates are from chain-propagated GBR counterfactual simulation. Each action is rated for evidence strength (High / Medium) based on cross-method sign agreement.

5.1 Process Setpoint Actions

Action	Specification	Est. Δ	Trade-off	Feasibility	Confidence
Extend cooling time	+1.5 s plantwide	−0.44 p.p.	+1.0% cycle time; negligible energy	Existing equip.	High

Cap mold temperature	≤ 78 °C	−0.21 p.p.	Negligible	Existing equip.	Med–High
Lower dryer dewpoint	−5 °C when humidity ≥ 65%	−0.09 p.p.	Small energy uptick	Existing equip.	Medium
Reduce injection pressure	−30 bar when wear_index ≥ 0.45	−0.09 p.p.	Monitor short-shot rate	Existing equip.	Medium

Table 4 — Direct process setpoint actions (real-time adjustments). Est. Δ = chain-propagated GBR counterfactual impact; p.p. = percentage points of scrap_rate_pct.

5.2 Maintenance Scheduling

Action	Specification	Est. Δ	Trade-off	Confidence
Preventive maintenance	Cap interval at 14 days	−0.07 p.p.	Labour cost only; no production interruption	Medium

Table 5 — Planning-level action. The reported direct β understates the benefit: the primary mechanism is indirect via reduced calibration_drift_index. Mediator cannot be in the adjustment set (see §2.1), so the total effect, flowing through drift, is what drives the recommendation.

5.3 Combined Package

Metric	Baseline	With package
Mean scrap rate	4.44%	≈ 3.85% (−13% relative)
Cycle time	53.7 s	≈ 54.3 s (+1.0%)
Energy per interval	19.01 kWh	≈ 19.02 kWh (+0.07%)

Table 6 — Simulated combined-package impact. The dominant trade-off is cycle time, driven almost entirely by the +1.5 s cooling extension. The scrap-to-cycle-time ratio is approximately 10:1 under the fitted model.

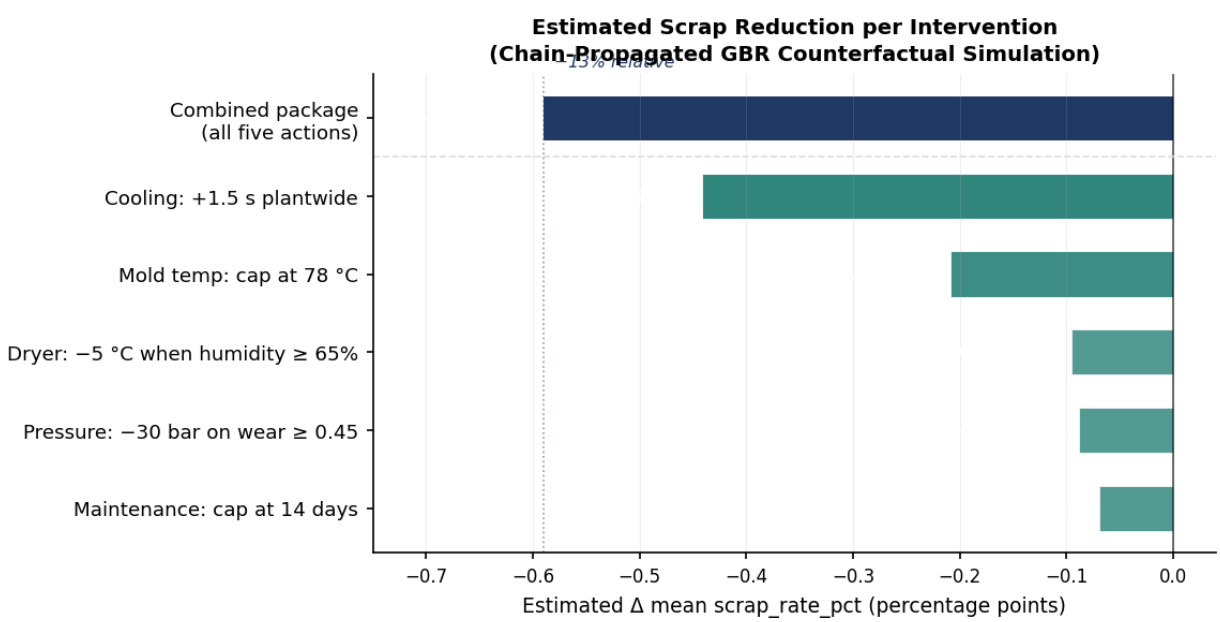

Figure 4 — Estimated scrap reduction per intervention, sorted by magnitude. Teal bars: individual actions. Dark blue bar: combined package. The dotted line at −0.59 p.p. marks the combined package estimate (−13% relative vs. baseline). Cooling time alone contributes approximately 75% of the package impact.

5.4 Deployment Sequence

Phase 1 (weeks 1–2): Roll out the three lowest-risk setpoint changes — cooling +1.5 s, mold-temperature cap at 78 °C, dryer trigger rule — and the 14-day maintenance cap. Pilot at VN_QUANGNAM (highest current scrap rate, highest humidity exposure). Expected path: 4.44% → ~3.95%.

Phase 2 (weeks 3–8): Introduce wear-aware pressure control: reduce injection pressure by 30 bar when tool_wear_index ≥ 0.45. Use short-shot defect rate as a safety gate — roll back if short-shot incidence rises more than 15% relative over any two-week window. Expected path: ~3.95% → ~3.85%.

Verification: Track weekly scrap by plant and machine; review defect-mix composition biweekly; target pass-fail rate below 55% at one month, below 50% at two months.

5.5 Actions Explicitly Not Recommended

- Shortening cooling time. Raw correlation invites this; DAG-adjusted analysis reverses it. Shortening cooling is estimated to raise scrap by ~0.41 p.p./s under the stated model.
- Plant-wide HVAC to control humidity. Humidity is a confounder — an environmental condition outside the intervention scope of the ontology. The correct lever is the dryer dewpoint.
- Reassigning experienced operators. The positive β on operator experience is attributable to residual assignment bias, not a direct causal effect. No intervention follows.
- Blanket injection-pressure reductions. On lightly worn tooling, elevated pressure is not associated with elevated scrap. The recommendation is strictly conditional on tool_wear_index ≥ 0.45.

6. Limitations

Residual confounding. Fixed effects absorb stable machine-, mold-, and shift-level heterogeneity but not time-varying unobserved factors such as undocumented mold changes, raw-material handling differences across batches, or production schedule pressure. The identification strategy assumes the DAG is correctly specified; unmeasured common causes not encoded in the graph would bias β_1. Sensitivity analyses varying the adjustment set confirm that the sign of the cooling-time effect is robust across plausible alternative specifications; the magnitude is not.

Moisture model weakness. The structural sub-model linking dryer_dewpoint_c to resin_moisture_pct explains only $R^2 = 0.09$ of moisture variance. The three measured parents (ambient humidity, dewpoint setting, batch ID, ambient temperature) are weak predictors at the interval level. The dryer recommendation is accordingly sized conservatively and should be treated as a lower-bound effect estimate; pilot data may reveal greater benefit if unobserved dryer mechanical health accounts for the missing variance.

Bootstrap inference. Confidence intervals are computed using a row-wise (i.i.d.) bootstrap. Adjacent 30-minute intervals on the same machine share equipment-level noise; within-machine observations are not independent. A machine-clustered bootstrap would widen the reported 95% intervals, overstating precision for magnitude estimates. Signs and rankings are robust to this issue; magnitudes should be read as approximate.

Temporal and geographic scope. The dataset covers 75 days in winter and early spring. Ambient humidity at the Vietnamese plants in summer may differ substantially; the humidity-conditional dryer rule is sized on in-sample humidity ranges only and may require recalibration for other seasons. No external validity claim is made beyond the Karcher group.

No cost data. All estimates are in scrap percentage points and seconds of cycle time. No monetary conversion is provided; any financial figures in accompanying materials should be replaced with plant-specific cost data before an investment decision is made.

Given these limitations, the two-week pilot at VN_QUANGNAM with defect-mix monitoring as a safety gate is not optional — it converts model-implied estimates into empirical evidence and limits the risk of acting on residual confounding.

7. Conclusion

Across 5,000 production intervals and 12 injection-molding machines, the dominant source of scrap is warpage driven by insufficient cooling relative to mold temperature, affecting one-third of all intervals. Flash and dimensional deviation contribute a further 37%, both sensitive to the interaction between tool wear state and pressure settings. Splay moisture, while lower in frequency, has the highest per-interval mean scrap rate and is partially addressable through the dryer dewpoint lever under elevated ambient humidity.

The central analytical finding is a sign change in the dataset's most important lever: the raw positive correlation between cooling time and scrap ($\rho = +0.28$) reflects reverse causation — operators compensating for thermal stress — while the DAG-adjusted estimate ($\beta = -1.75$ standardized; $\tilde{\beta} = -0.41$ p.p./s) reveals a protective effect. A predictive pipeline acting on the raw signal would produce the wrong recommendation. This sign change is the clearest demonstration that causal structure — encoded in the supplied DAG — is not a methodological addition but a practical necessity for operational decision-making.

The five-action package — cooling +1.5 s, mold temperature capped at 78 °C, dryer dewpoint −5 °C under high humidity, injection pressure −30 bar on worn tooling, maintenance capped at 14-day intervals — is estimated to reduce mean scrap from 4.44% to approximately 3.85% (−13% relative), at a cycle-time cost of ~1% and negligible energy impact. All estimates are model-implied from observational data. A two-week controlled pilot at VN_QUANGNAM, with defect-mix composition as a safety gate, is the recommended next step before plant-wide deployment.

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